So, with our newfound power to analyze string theory, what insights have emerged? There have been many. I will focus on those that have had the greatest impact on the story of space and time.

Of primary importance, Witten's work revealed that the approximate string theory equations used in the 1970s and 1980s to conclude that the universe must have nine space dimensions *missed the true number by one*. The exact answer, his analysis showed, is that the universe according to M-theory has ten space dimensions, that is, eleven spacetime dimensions. Much as Kaluza found that a universe with five spacetime dimensions provided a framework for unifying electromagnetism and gravity, and much as string theorists found that a universe with ten spacetime dimensions provided a framework for unifying quantum mechanics and general relativity, Witten found that a universe with eleven spacetime dimensions provided a framework for unifying all string theories. Like five villages that appear, viewed from ground level, to be completely separate but, when viewed from a mountaintop—making use of an additional, vertical dimension—are seen to be connected by a web of paths and roadways, the additional space dimension emerging from Witten's analysis was crucial to his finding connections between all five string theories.

While Witten's discovery surely fit the historical pattern of achieving unity through more dimensions, when he announced the result at the annual international string theory conference in 1995, it shook the foundations of the field. Researchers, including me, had thought long and hard about the approximate equations being used, and everyone was confident that the analyses had given the final word on the number of dimensions. But Witten revealed something startling.

He showed that all of the previous analyses had made a mathematical simplification tantamount to *assuming* that a hitherto unrecognized tenth spatial dimension would be extremely small, much smaller than all others. So small, in fact, that the approximate string theory equations that all researchers were using lacked the resolving power to reveal even a mathematical hint of the dimension's existence. And that led everyone to conclude that string theory had only nine space dimensions. But with the new insights of the unified M-theoretic framework, Witten was able to go beyond the approximate equations, probe more finely, and